

Thu, Oct. 30

Exam 2

8:00 - 9:15

Arrive at least 5 min early!

Stepan Center

(Only 10 (best) counts)

Format:

11 multiple choice questions

3 free response questions

Practice exams on Canvas



The exam will cover:

Chapter 14
and Sections 15.1, 15.2

Lectures
11 - 22

See also
Review 4 & 5

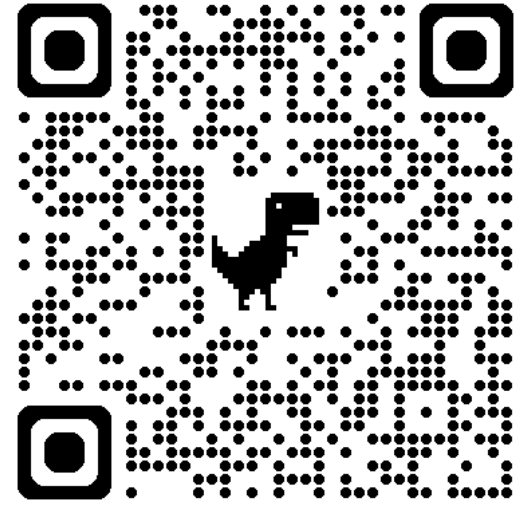
Good
luck!



Calculators: NOT allowed

Notes:

- Limits and Continuity
- Chain Rule (and Implicit differentiation)
- Review 4
- Review 5
- Lectures 11-22



Functions of Several Variables and level curves (Lecture 11)
Partial derivatives (Lectures 12, 14)
Directional derivatives and gradient (Lect. 15, 16) } Review 4

Local Extrema (Lect. 17)
Max/min on bounded regions (Lect. 18)
Lagrange multipliers (Lect. 19, 20)
Double integrals (Lect. 21, 22) } Review 5

1) What is the geometrical meaning of

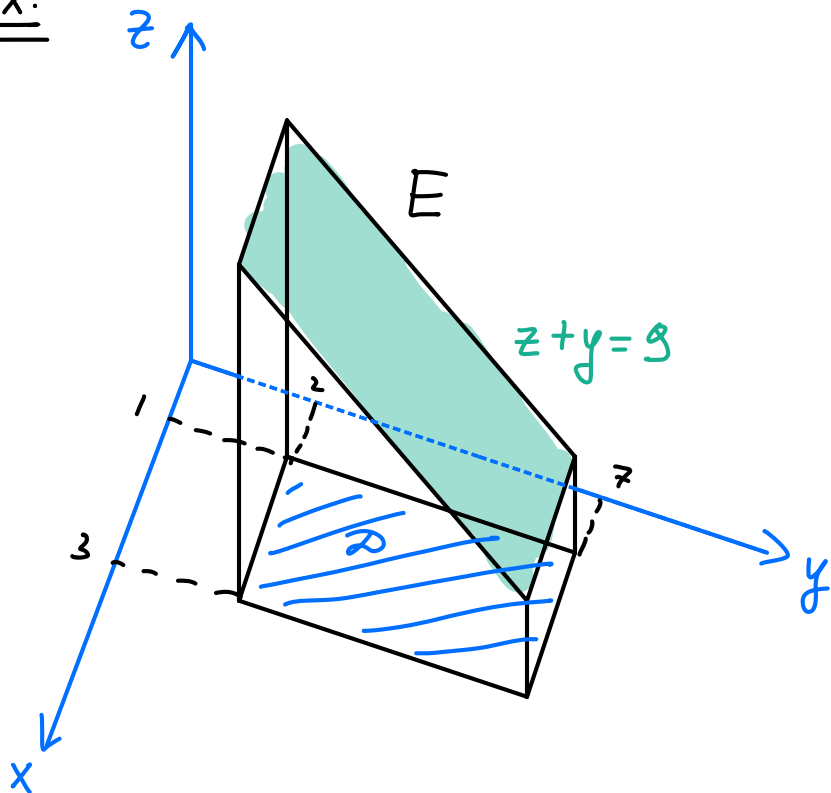
$$\iint_{\mathcal{D}} 1 \, dA?$$

A: $\iint_{\mathcal{D}} 1 \, dA = \text{Area}(\mathcal{D})$

2) If $f \geq 0$ what is the geometrical meaning of

$$\iint_{\mathcal{D}} f(x,y) \, dA?$$

Ex:



Find the volume of E .

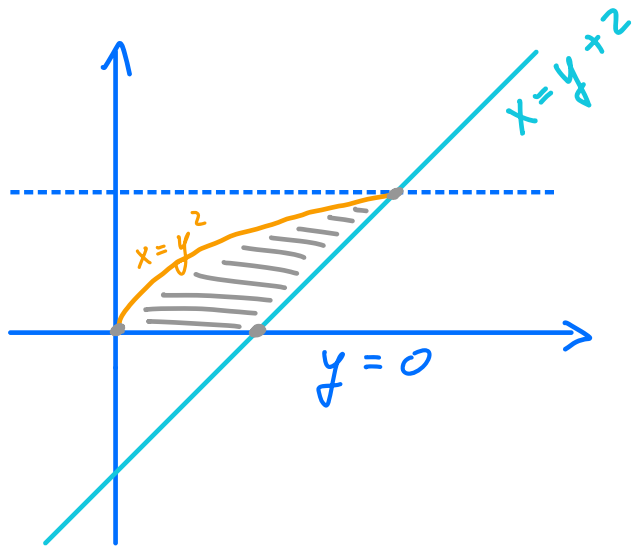
$$\iint_{\mathcal{D}} f(x,y) \, dA = \int_1^3 \int_2^7 (9-y) \, dy \, dx$$

$$\parallel z + y = 9 \Leftrightarrow z = 9 - y \quad f(x,y) = 9 - y \parallel$$

$$\text{Volume}(E) = \int_1^3 \int_2^7 (9-y) \, dy \, dx$$

Ex: Let D be a region bounded by $y=0$ on the bottom, $x=y^2$ on top and $x=y+2$ on the right. Rewrite $\iint_D F(x,y) dA$ as an iterated integral.

Sol:



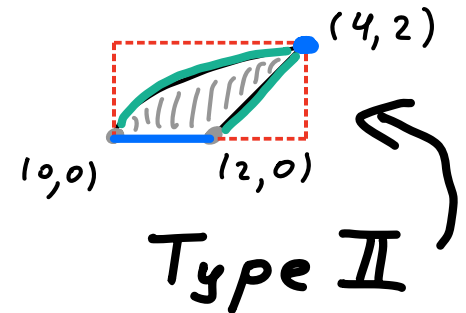
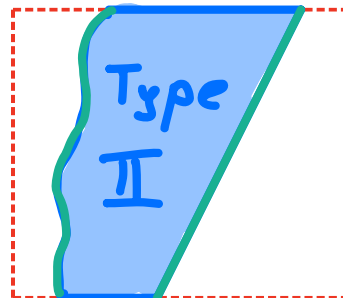
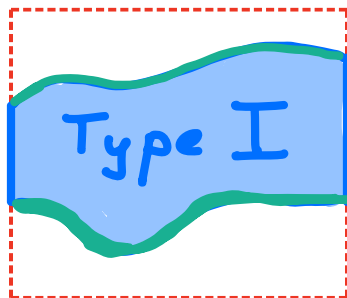
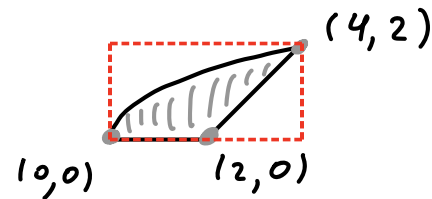
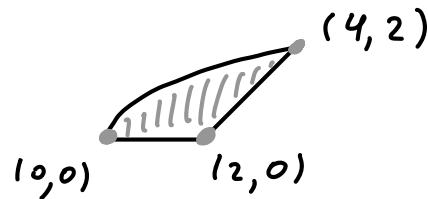
1) Sketch D

$$\begin{cases} y=0 \\ x=y+2 \end{cases} \quad (2,0)$$

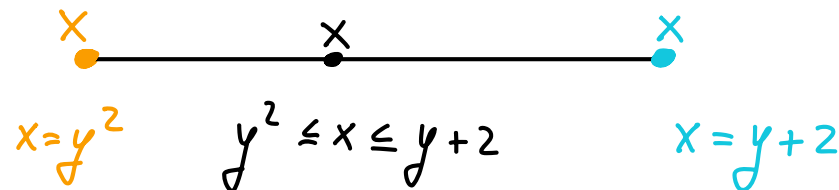
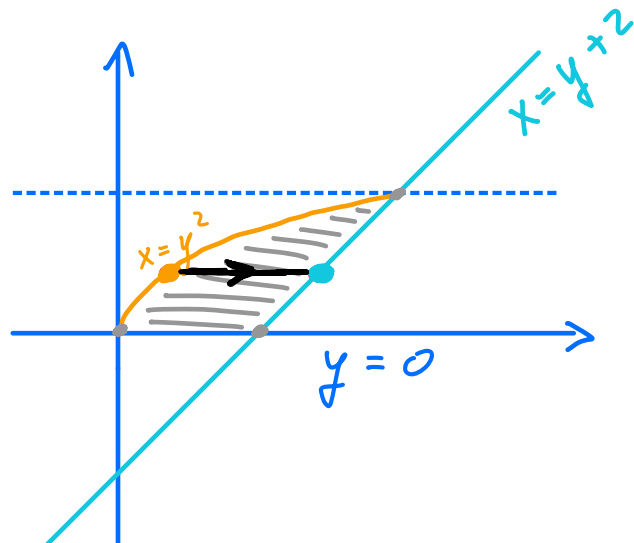
$$\begin{cases} x=y^2 \\ y=0 \end{cases} \quad (0,0)$$

$$\begin{cases} x=y+2 \\ x=y^2 \end{cases} \text{ \& } y \geq 0 \quad (4,2)$$

2) Type I vs Type II

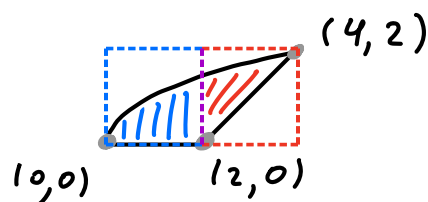
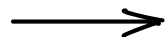
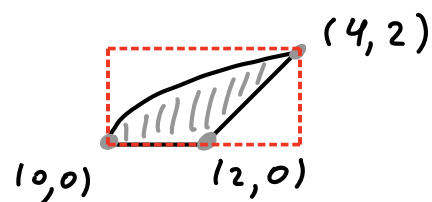


$$3) \mathcal{D} = \{ (x, y) \mid 0 \leq y \leq 2, y^2 \leq x \leq y+2 \}$$

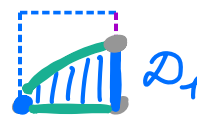


$$4) \iint_{\mathcal{D}} f(x, y) dA = \int_0^2 \int_{y^2}^{y+2} f(x, y) dx dy$$

Q: Could we change the order of integration? Yes!



$$\mathcal{D} = \mathcal{D}_1 \cup \mathcal{D}_2$$



$$\mathcal{D}_1: \begin{aligned} 0 &\leq x \leq 2 \\ 0 &\leq y \leq \sqrt{x} \end{aligned}$$

$$\mathcal{D}_2: \begin{aligned} 2 &\leq x \leq 4 \\ x-2 &\leq y \leq \sqrt{x} \end{aligned}$$

$$\int_0^2 \int_{y^2}^{y+2} f(x, y) dx dy = \int_0^2 \int_0^{\sqrt{x}} f(x, y) dy dx + \int_2^4 \int_{x-2}^{\sqrt{x}} f(x, y) dy dx$$

For more examples
see Review 5

1) $F(x, y, z)$ $\nabla F = \langle F_x, F_y \rangle$ (F)

$$\nabla F(x, y) = \langle F_x(x, y), F_y(x, y) \rangle$$

$$\nabla h(x, y, z) = \langle h_x(x, y, z), h_y(x, y, z), h_z(x, y, z) \rangle$$

2) $F(x, y) = e^x + xy$, $\vec{v} = \langle 3, 4 \rangle$ then the directional derivative of F in the direction of $\vec{v}$ is $3(e^x + y) + 4x$ (F)

$$\vec{u} = \frac{\vec{v}}{|\vec{v}|} = \langle \frac{3}{5}, \frac{4}{5} \rangle \quad \nabla F = \langle F_x, F_y \rangle = \langle e^x + y, x \rangle \quad \mathcal{D}_{\vec{u}} F = \vec{u} \cdot \nabla F = \frac{3}{5}(e^x + y) + \frac{4}{5}x$$

3) $\nabla F(x, y)$ is perpendicular to the surface $z = F(x, y)$ (F)

$$F(x, y, z) = F(x, y) - z \quad F(x, y, z) = 0 \Leftrightarrow z = F(x, y)$$

$$\nabla F = \langle F_x, F_y, -1 \rangle \text{ perp. to level surf.}$$

4) What is the direction of max decrease of $F(x, y, z)$?

$$-\nabla F = \langle -F_x, -F_y, -F_z \rangle \text{ with the rate } |\nabla F|.$$

5) How to find tangent plane/normal line to $z = F(x, y)$?

$$F(x, y, z) = F(x, y) - z, \text{ consider level surface } F(x, y, z) = 0$$

∇F is a direction vector for the line and normal vector for the plane.

True-False

Ex: Show that $\lim_{(x,y) \rightarrow (1,1)} \frac{(x-1)^2}{(y-1)^2}$ does NOT exist.

Sol:

$$F(x,y) = \frac{(x-1)^2}{(y-1)^2}$$

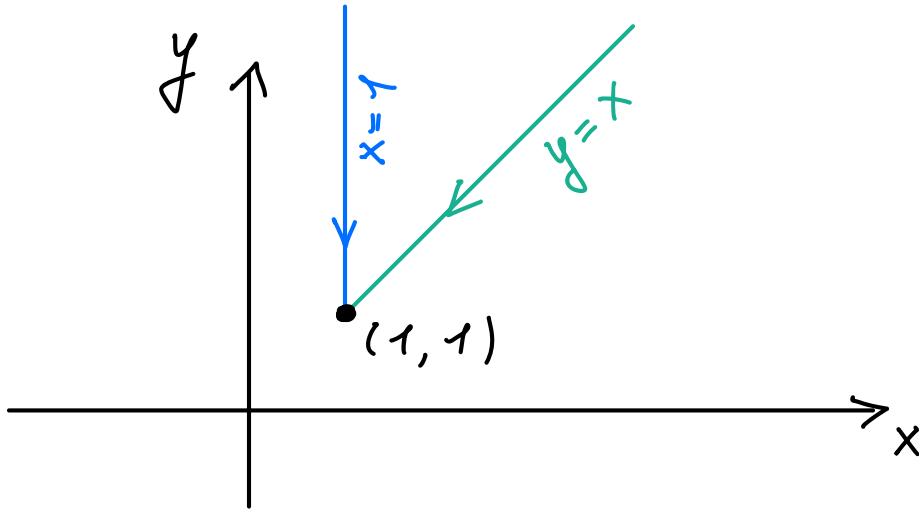
Approaching along $y=x$:

$$F(x,x) = \frac{(x-1)^2}{(x-1)^2} = 1$$

Approaching along $x=1$:

$$F(1,y) = \frac{(1-1)^2}{(y-1)^2} = 0$$

$0 \neq 1 \Rightarrow$ the limit does NOT exist.



$f(x,y)$ - continuous function.

1) What can we say about $f(x_0, y_0)$ if $\nabla f(x_0, y_0) = 0$?
 (x_0, y_0) is a critical point,
 f has a local max/min or a saddle point at (x_0, y_0)

2) What is the difference between

(Find max/min values of $f(x,y)$
subject to constraint $g(x,y)=k$) AND (Find max/min of $f(x,y)$
on bounded regions)?

Ex: [Find extreme values of $f(x,y) = x^2 + 2y^2$
on the circle $x^2 + y^2 = 1$.] ← subject to constraint

max/min on → [Find extreme values of $f(x,y) = x^2 + 2y^2$
bounded region on the disk $x^2 + y^2 \leq 1$]

See Lecture 19 for solutions of both.

On the plane

- $\underbrace{g(x,y)=k}$ is a level curve of the function g
 $\uparrow$ describes a CURVE

- the boundary of a region is a curve

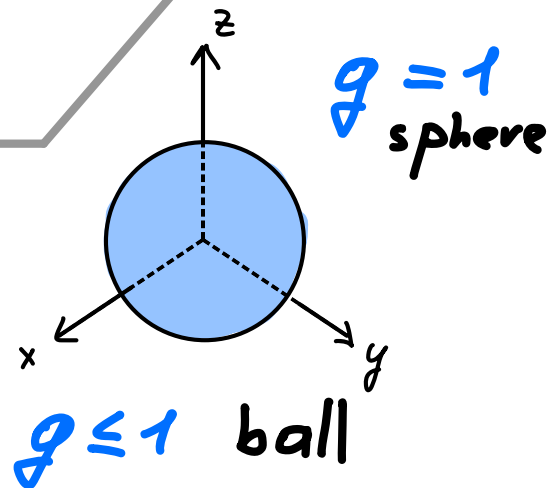
Ex: $g(x,y) = x^2 + y^2$



In the space

- $g(x,y,z)=k$ is a level surface
- the boundary of a solid is a surface

Ex: $g(x,y,z) = x^2 + y^2 + z^2$

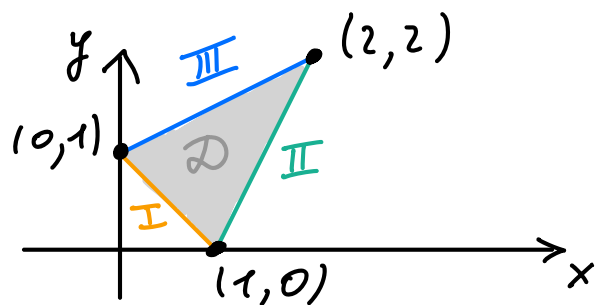


1) Find the points on the sphere $x^2 + y^2 + z^2 = 4$ closest to / farthest from the point $P(3, 1, -1)$

2) The room has triangular shape with vertices at $(0, 1)$, $(1, 0)$ and $(2, 2)$. The temperature is given by $t(x, y) = xy - 2x^2$. Where is the coolest place in the room?

(1) - subject to constraint
[see Lecture 18]

(2) - min in the region



Step 1: Find critical point inside $\mathcal{D}$

$$\nabla t = \langle y - 4x, x \rangle \quad \begin{cases} y - 4x = 0 \\ x = 0 \end{cases} \quad \begin{cases} x = 0 \\ y = 0 \end{cases}$$

$(0, 0)$ is NOT inside $\mathcal{D}$

Step 2: Find min values on the boundary

In order to do step 2 divide the boundary into 3 pieces and find min value at each piece of the boundary:

$$x + y = 1$$

$$2x - y = 2$$

$$2y - x = 2$$

$$\text{I) } y = 1 - x, \quad 0 \leq x \leq 1$$

$$\begin{cases} f(x, y) = f(x, 1-x) = x(1-x) - 2x^2 \\ 0 \leq x \leq 1 \end{cases}$$

$$\begin{cases} f(x, 1-x) = x - 3x^2 \\ 0 \leq x \leq 1 \end{cases}$$

WANTED: min of $h(x) = x - 3x^2$ on $[0, 1]$.

$$h'(x) = 1 - 6x \quad h'(\frac{1}{6}) = 0$$

$$h(0) = 0$$

$$h(1) = -2 \quad \leftarrow \text{min}$$

$$h(\frac{1}{6}) = \frac{1}{12}$$

$$f(1, 1-1) = f(1, 0) = -2$$

$$\text{II) } y = 2x - 2, \quad 1 \leq x \leq 2$$

$$\begin{cases} f(x, y) = f(x, 2x-2) = x(2x-2) - 2x^2 \\ 1 \leq x \leq 2 \end{cases}$$

$$\begin{cases} f(x, 2x-2) = -2x \\ 1 \leq x \leq 2 \end{cases}$$

WANTED: min of $h(x) = -2x$ on $[1, 2]$

$$h(1) = -2 \quad h(2) = -4 \quad \leftarrow \text{min}$$

$$f(2, 2 \cdot 2 - 2) = f(2, 2) = -4$$

$$\text{III) } y = \frac{1}{2}(2+x), \quad 0 \leq x \leq 2$$

$$\begin{cases} t(x, y) = t(x, \frac{1}{2}(2+x)) = x \frac{2+x}{2} - 2x^2 \\ 0 \leq x \leq 2 \end{cases}$$

$$\begin{cases} t(x, \frac{2+x}{2}) = x - \frac{3}{2}x^2 \\ 0 \leq x \leq 2 \end{cases}$$

WANTED: min of $h(x) = x - \frac{3}{2}x^2$
on $[0, 2]$

$$h'(x) = 1 - 3x \quad h'(\frac{1}{3}) = 0$$

$$h(0) = 0$$

$$h(2) = -4 \quad \leftarrow \text{min}$$

$$h(\frac{1}{3}) = \frac{1}{6}$$

$$t(2, \frac{2+2}{2}) = t(2, 2) = -4$$

The smallest temperature is at the point $(2, 2)$.

Find max/min of $f(x,y,z)$ subject to two constraints
 $g(x,y,z)=k$ $h(x,y,z)=p$

$$* \begin{cases} f_x = \lambda g_x + \mu h_x \\ f_y = \lambda g_y + \mu h_y \\ f_z = \lambda g_z + \mu h_z \\ g = k \\ h = p \end{cases}$$

Goal: we want to solve (*)

step 1: Think about λ and μ as GIVEN and solve the system

$$\begin{cases} f_x = \lambda g_x + \mu h_x \\ f_y = \lambda g_y + \mu h_y \\ f_z = \lambda g_z + \mu h_z \end{cases}$$

Here you have 3 equations and 3 variables (x,y,z)

So you should be able to write each of x,y and z in terms of λ and μ .

step 2: Use expressions you get in step 1 to solve

$$\begin{cases} g(x,y,z)=k \\ h(x,y,z)=p \end{cases}$$

Here you get 2 equations and 2 variables (λ and μ).

step 3: Combine steps 1 & 2.

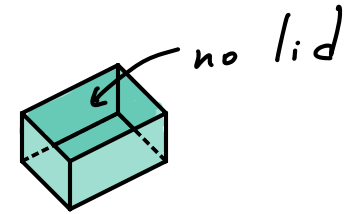
Rmk: This is A way to solve (*), not THE.
(see the next page)

Ex: Rectangular box without a lid is made of 12m^2 of cardboard.
What is the max possible volume?

Sol: $x, y, z > 0$ $f(x, y, z) = xyz$ - volume
 $g(x, y, z) = \underbrace{xy + 2xz + 2yz}_{\text{surface area}} = 12$

single constraint!

$$\begin{cases} yz = \lambda(y + 2z) & \xrightarrow{\cdot \frac{1}{yz}} & 1 = \lambda\left(\frac{1}{z} + \frac{2}{y}\right) & \textcircled{1} \\ xz = \lambda(x + 2z) & \xrightarrow{\cdot \frac{1}{xz}} & 1 = \lambda\left(\frac{1}{z} + \frac{2}{x}\right) & \textcircled{2} \\ xy = \lambda(2x + 2y) & \xrightarrow{\cdot \frac{1}{xy}} & 1 = \lambda\left(\frac{2}{y} + \frac{2}{x}\right) & \textcircled{3} \\ xy + 2xz + 2yz = 12 \end{cases}$$



Note that $\lambda \neq 0$

$$\textcircled{1} - \textcircled{2}: \quad 2\lambda\left(\frac{1}{y} - \frac{1}{x}\right) = 0 \quad \Rightarrow \quad y = x \quad 1 = \lambda\left(\frac{1}{z} + \frac{2}{x}\right) \Rightarrow 1 = \lambda \frac{4}{x} \Rightarrow \lambda = \frac{x}{4}$$

$$\textcircled{1} - \textcircled{3}: \quad \lambda\left(\frac{1}{z} - \frac{2}{x}\right) = 0 \quad \Rightarrow \quad z = \frac{x}{2} \quad xy + 2xz + 2yz = 12 \Rightarrow 3x^2 = 12$$

$$\Rightarrow x = \pm 2 \\ \text{should be } > 0 \Rightarrow 2$$

$$\boxed{\begin{aligned} x &= 2, \quad y = 2, \quad z = 1 \\ \text{max volume: } & 2 \cdot 2 \cdot 1 = \underline{\underline{4}} \end{aligned}}$$

Ex: Find min of $f(x,y) = x$ subject to constraint $y^2 + 4x^2 = 1$.

Sol: $g(x,y) = y^2 + 4x^2$

$$\nabla f = \langle 1, 0 \rangle$$

$$\nabla g = \langle 8x, 2y \rangle$$

$$\begin{cases} \nabla f = \lambda \nabla g \\ g(x,y) = 1 \end{cases}$$

1) IF $\lambda \neq 0$

$$\begin{cases} 1 = \lambda(8x) \\ 0 = \lambda(2y) \\ y^2 + 4x^2 = 1 \end{cases}$$

$$\begin{cases} x = \frac{1}{8\lambda} \\ y = 0 \\ y^2 + 4x^2 = 1 \end{cases} \Leftrightarrow \begin{cases} x = \frac{1}{8\lambda} \\ y = 0 \\ 4\left(\frac{1}{8\lambda}\right)^2 = 1 \end{cases}$$

$$\parallel \lambda = \pm \frac{1}{4} \parallel$$

$$\lambda = \frac{1}{4} \Rightarrow x = \frac{1}{2}, y = 0$$

$$f\left(\frac{1}{2}, 0\right) = \frac{1}{2}$$

$$\lambda = -\frac{1}{4} \Rightarrow x = -\frac{1}{2}, y = 0$$

$$f\left(-\frac{1}{2}, 0\right) = -\frac{1}{2}$$

2) IF $\lambda = 0$

$$\begin{cases} 1 = 0(8x) \\ 0 = 0(2y) \\ y^2 + 4x^2 = 1 \end{cases} \leftarrow \text{Never holds} \Rightarrow \lambda \neq 0$$

The minimum is at the point $\left(-\frac{1}{2}, 0\right)$ and equal to $-\frac{1}{2}$.